Assignment 4: Bayesian Inference, Conjugate Priors, and Probabilistic Graphical Models

Math4AI: Probability & Statistics

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1 Introduction

Bayesian statistics provides a robust framework for updating beliefs in the presence of new data. Unlike frequentist approaches, Bayesian inference treats parameters as random variables with associated probability distributions. This report covers the Beta-Binomial conjugate model, the trade-offs between Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP), and the exact inference in Bayesian Networks using joint distribution marginalization.

2 Mathematical Foundations of Bayesian Inference

2.1 The Beta-Binomial Conjugate Model

In Bernoulli trials with an unknown probability of success θ, the Binomial distribution serves as the likelihood function. For Bayesian updating, we use the Beta distribution as a prior due to its property as a **conjugate prior**. The prior density is defined as:

$$P(\theta|\alpha, \beta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)} \tag{1}$$

where $B(\alpha, \beta)$ is the Beta function. When we observe h heads and t tails, the posterior distribution follows:

$$P(\theta|\text{Data}) \propto P(\text{Data}|\theta)P(\theta) \propto \theta^h(1-\theta)^t \cdot \theta^{\alpha-1}(1-\theta)^{\beta-1} \tag{2}$$

This results in a posterior Beta distribution: $\text{Beta}(\alpha + h, \beta + t)$. This "closed-form" update is the cornerstone of efficient Bayesian computation.

2.2 Point Estimation: MLE vs. MAP

While the posterior provides a full distribution, we often require a single point estimate:

1. **Maximum Likelihood Estimation (MLE):** Ignores the prior and maximizes the likelihood:
$$\hat{\theta}_{MLE} = \frac{\text{heads}}{\text{heads} + \text{tails}} \tag{3}$$

2. **Maximum A Posteriori (MAP):** Incorporates the prior and finds the mode of the posterior. For $\alpha, \beta > 1$:
$$\hat{\theta}_{MAP} = \frac{\alpha + h - 1}{\alpha + \beta + h + t - 2} \tag{4}$$

MAP acts as a **regularizer**. In cases with small data (e.g., 5 heads, 0 tails), MLE yields an extreme 100% probability, whereas MAP pulls the estimate toward the prior, preventing overfitting to sparse observations.

2.3 Bayesian Networks and Exact Inference

A Bayesian Network represents a factorized joint probability distribution over a set of variables $\{X_1, \ldots, X_n\}$. Based on the network topology (Rain $\rightarrow$ Sprinkler $\rightarrow$ Wet Grass), the joint distribution is:

$$P(R, S, W) = P(R) \cdot P(S|R) \cdot P(W|S, R) \tag{5}$$

To answer a conditional query $P(Q|E)$, where E is the observed evidence, we perform **marginalization** over the hidden variables H:

$$P(Q|E) = \frac{P(Q, E)}{P(E)} = \frac{\sum_{h \in H} P(Q, E, h)}{\sum_{q,h} P(q, E, h)} \tag{6}$$

This approach, while computationally expensive for large networks, provides exact probabilistic answers by summing consistent states in the joint probability table.

3 Verification and Simulation Results

3.1 Evolution of Belief (Beta Update Plot)

The following visualization shows how the agent's belief about θ (the probability of heads) shifts and sharpens as more data is observed. Starting from a uniform Beta$(1, 1)$ prior, the density concentrates around the true $\theta_{true} = 0.8$.

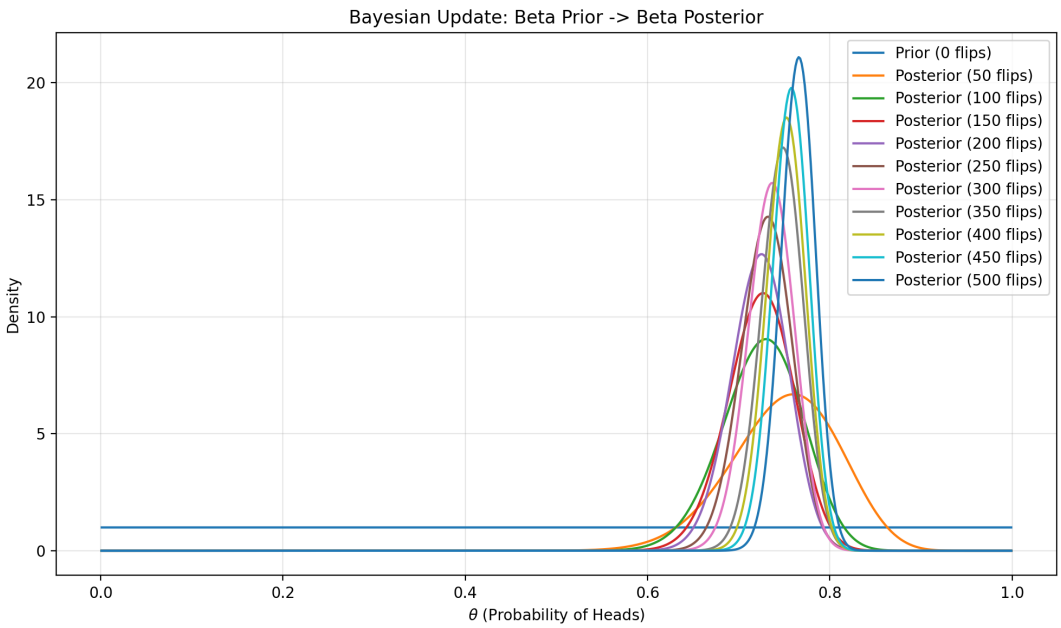

Figure 1: The sharpening of the posterior distribution over 500 coin flips.

3.2 MLE vs. MAP Comparison

In the simulation with $h = 5$ and $t = 0$ using a Beta$(10, 10)$ prior, the results were:

- **MLE:** 1.0000 (Over-confident due to lack of prior)
- **MAP:** 0.6087 (Regularized by the prior belief)

This confirms that the Bayesian approach is more robust to "zero-frequency" problems.

3.3 Wet Grass Network Inference

The exact inference query for the probability of Rain given that the grass is wet yielded:

$$P(R = True|W = True) \approx 0.3577 \tag{7}$$

The top-3 joint assignments highlight that the most probable state is $(False, False, False)$—no rain, no sprinkler, and dry grass—accounting for approximately 48% of the total probability space.

4 Conclusion

The Bayesian framework successfully updated beliefs through conjugate priors and provided regularized estimates that outperformed simple MLE in sparse data scenarios. Furthermore, exact inference through marginalization demonstrated the power of joint probability factorization in causal networks.

A Appendix: Implementation Logic

```
# Bayesian Posterior Update
self.alpha += heads
self.beta += tails

# Joint Distribution Marginalization
for assignment, prob in self.joint.items():
    if matches_evidence(assignment, evidence):
        denominator += prob
        if assignment[q_idx] == True:
            numerator += prob
```